

Day 2 – Python Loops (While Loop Detailed Lecture)

1. What is a Loop?

A loop is used to repeat a block of code multiple times automatically.

2. Types of Loops

1. While Loop
2. For Loop

3. While Loop Syntax

```
while condition:  
    # code block
```

4. Example

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1
```

5. Important Concept

Always update the loop variable, otherwise it will become an infinite loop.

6. Program 1: Sum of N Numbers

Logic: Add numbers from 1 to N using a loop.

7. Program 2: Even Numbers

Logic: Use modulus (%) to check even numbers.

8. Program 3: Reverse Counting

Logic: Start from N and decrease until 1.

9. Common Mistakes

- Forgetting increment
- Wrong condition
- Infinite loops

10. Summary

You learned while loop, repetition logic, and basic loop programs.